DCAT 3 - ontology

http://www.w3.org/ns/dcat#

Maintained by W3C

In [ ]:
```
# parse DCAT as RDF graph
import rdflib

prefixes = '''
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX skos: <http://www.w3.org/2004/02/skos/core#>
PREFIX dct: <http://purl.org/dc/terms/>
PREFIX dcat: <http://www.w3.org/ns/dcat#>
'''
dcat_g = rdflib.Graph()
dcat_g.parse("http://www.w3.org/ns/dcat#")
```

Out[ ]:
```
<Graph identifier=N4a9d270148f041008e1d4d49427aeead (<class 'rdflib.graph.G
raph'>)>
```

In [28]:
```
# labels and definitions of classes:     dcat:DataService, dcat:Dataset, dca

q_dcat_Dataset =  prefixes + '''
CONSTRUCT{
    ?subj a ?class ;
          rdfs:label ?label ;
          skos:definition ?def ;
          skos:scopeNote ?scope ;
}
WHERE {
  VALUES ?subj {
    dcat:DataService
    dcat:Dataset
    dcat:Distribution
      }
    ?subj a ?class ;
          skos:definition ?def ;
          rdfs:label ?label ;
    FILTER( LANG(?def) = 'en'&&  LANG(?label) = 'en')
}
'''
```

In [30]:
```
qr = dcat_g.query(q_dcat_Dataset)
construct_graph = qr.graph
print(str(construct_graph.serialize()))
```

```
@prefix dcat: <http://www.w3.org/ns/dcat#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .

dcat:DataService a rdfs:Class,
        owl:Class ;
    rdfs:label "Data service"@en ;
    skos:definition "A site or end-point providing operations related to the
discovery of, access to, or processing functions on, data or related resourc
es."@en .

dcat:Dataset a rdfs:Class,
        owl:Class ;
    rdfs:label "Dataset"@en ;
    skos:definition "A collection of data, published or curated by a single
source, and available for access or download in one or more representation
s."@en .

dcat:Distribution a rdfs:Class,
        owl:Class ;
    rdfs:label "Distribution"@en ;
    skos:definition "A specific representation of a dataset. A dataset might
be available in multiple serializations that may differ in various ways, inc
luding natural language, media-type or format, schematic organization, tempo
ral and spatial resolution, level of detail or profiles (which might specify
any or all of the above)."@en .
```

DCAT-AP 3.0.0

Maintained by the SEMIC Support Centre.

> DCAT-AP is a DCAT profile for sharing information about Catalogues containing Datasets and Data Services descriptions in Europe, under maintenance by the SEMIC action, Interoperable Europe.

> This Application Profile provides a minimal common basis within Europe to share Datasets and Data Services cross-border and cross-domain. https://semiceu.github.io/DCAT-AP/releases/3.0.0/

> DCAT-AP allows:
>
> - Data catalogues to describe their dataset collections using a standardised description, while keeping their own system for documenting and storing them.
> - Content aggregators, such as the European Data Portal, to aggregate such descriptions into a single point of access.

> - Data consumers to more easily find datasets through a single point of access.

https://interoperable-europe.ec.europa.eu/collection/semic-support-centre/solution/dcat-application-profile-data-portals-europe

DCAT-AP purpose of facilitatting the reuse of data, in concrete:

> - It proposes mandatory, recommended or optional classes and properties to be used for a particular application;
> - It identifies requirements to control vocabularies for this particular application;
> - It gathers other elements to be considered as priorities or requirements for an application such as a conformance statement.

https://interoperable-europe.ec.europa.eu/collection/semic-support-centre/solution/dcat-application-profile-data-portals-europe

dcat-ap-overview

The `dcat:Dataset` in DCAT-AP

Property	URI	Definition	Reuse	Optionality
description	dct:description	A free-text account of the Dataset	E	mandatory
title	dct:title	A name given to the Dataset	E	mandatory
applicable legislation	dcatap:applicableLegislation	The legislation that mandates the creation or management of the Dataset	P	optional
documentation	foaf:documentation	A page or document about this Dataset	P	optional
has version	dcat:hasVersion	A related Dataset that is a version, edition, or adaptation of the described Dataset	P	optional
provenance	dct:provenance	A statement about the lineage of a Dataset	P	optional

Property	URI	Definition	Reuse	Optionality
sample	adms:sample	A sample distribution of the dataset	P	optional
source	dct:source	A related Dataset from which the described Dataset is derived	P	optional
version notes	adms:versionNotes	A description of the differences between this version and a previous version of the Dataset	P	optional

`dcat:Catalogue` in DCAT-AP

Property	Definition	Reuse	Optionality
description	A free-text account of the Catalogue	E	mandatory
publisher	responsible for making the Catalogue available	E	mandatory
title	A name given to the Catalogue	E	mandatory
applicable legislation	The legislation that mandates the creation or management of the Catalogue	P	optional

`dcat:Distribution` in DCAT-AP

TODO:

EU Recommendations: **TODO: reduce to a few quotes**

In 2013, the European Union revised some legislation related to the re-use of public data. The revision included the adoption of the "open by default" principle, which formulates that data of public bodies should be open and free by default and design, as well as easily accessible via API if possible. As a response to this new standard, European public administrations set up Open Data portals, setting the first steps to a European open data ecosystem. Lack of standardization lead to a fragmented landscape of in which it was difficult to exchange metadata between the over 150 data portals, and which all had to be queried individually.

The European Commission recognized the need to improve the fragmented open data environment in Europe. A common metadata language would make it easier to discover and reuse datasets across portals. With this in mind, the commission extended the already existing DCAT by developing additional requirements and recommendations that would adhere to the European context, resulting in **DCAT-AP** (DCAT Application Profile). It's a specification of DCAT that reuses terms but adds more specificity. It identifies more mandatory, recommended and optional elements that should be used in particular situations, as well as recommendations for controlled vocabularies that could be used. DCAT-AP is extendable, meaning that further needs for specifications can be implemented. Although it was developed in the context of Open Data portals in Europe, it can be used for any type of dataset.

DCAT-AP Extensions

[DCAT-AP for High-Value Datasets (HVD)]

> DCAT-AP HVD offers users of DCAT-AP the possibility to adhere to the High-Value Datasets Implementing Regulation (HVD IR) with little additional effort. The HVD IR was adopted by the European Commission in December 2022 as part of their data strategy and the growing importance of data, in particular for datasets that have been identified as being of high value.

[GeoDCAT-AP]

> GeoDCAT-AP makes it easier to share descriptions of spatial datasets between spatial data portals and general data portals, and thus helps increase public and cross-sector access to such datasets.

[StatDCAT-AP]

> StatDCAT-AP delivers specifications and tools that enhance interoperability between descriptions of statistical data sets within the statistical domain and between statistical data and open data portals.

In [10]:

```
from SPARQLWrapper import SPARQLWrapper, TURTLE, JSON

prefixes = '''
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX aat: <http://vocab.getty.edu/aat/>
PREFIX gvp: <http://vocab.getty.edu/ontology#>
PREFIX gvp_lang: <http://vocab.getty.edu/language/>
PREFIX skos: <http://www.w3.org/2004/02/skos/core#>
PREFIX skosxl: <http://www.w3.org/2008/05/skos-xl#>
PREFIX dcterms: <http://purl.org/dc/terms/>
'''

def sparql_query(query, endpoint, results_f):
```

```
    '''
    function
    * receives SPARQL query as input
    * runs the query against Getty Vocabularies SPARQL endpoint
    * returns the results in turtle (ttl) format
    '''
    sparql = SPARQLWrapper(endpoint)
    query = prefixes + query
    sparql.setQuery(query)
    sparql.setReturnFormat(results_f)
    results = sparql.query().convert()
    return results
```

In [ ]:

```
from datetime import date
today = date.today().strftime("%Y-%m-%d")
getty_concepts_construct = '''
CONSTRUCT {
    <http://vocabularies.dans.knaw.nl/aatconcepts> a skos:ConceptScheme ;
        dct:title "The Art and Architecture Thesaurus Concepts"@en ;
        rdfs:label "The Art and Architecture Thesaurus Concepts"@en ;
        rdfs:comment "The Art and Architecture Thesaurus Concepts (AATC) is
        dct:creator <https://ror.org/008pnp284> , <https://orcid.org/0000-00
        dct:created "%s"^^xsd:date ;
        dcterms:license <http://opendatacommons.org/licenses/by/1.0/> .

    ?concept a skos:Concept ;
        skos:inScheme <http://vocabularies.dans.knaw.nl/aatconcepts> ;
        skos:prefLabel ?label_literal_en ;
        skos:prefLabel ?label_literal_nl ;
        dcterms:issued ?issuedDate .
}

WHERE {
    ?concept a gvp:Concept ;
        skos:inScheme aat: ;
        skosxl:prefLabel ?preflabel .

    OPTIONAL {?concept dcterms:issued ?issuedDate .}

    { ?preflabel  dcterms:language aat:300388277 ; skosxl:literalForm ?label

    UNION

    { ?preflabel  dcterms:language aat:300387822 ; skosxl:literalForm ?label
      BIND (STRLANG(STR(?label_literal_en_us), 'en') AS  ?label_literal_en)}

    UNION

    { ?preflabel  dcterms:language aat:300388256 ; skosxl:literalForm ?label
    FILTER( STRSTARTS(str(?concept), str(aat:)) )
}
ORDER BY ?concept
LIMIT 10
'''  % today
```

```
# query_r = sparql_query(query=q_dcat_classes, endpoint="http://www.w3.org/r
query_r = sparql_query(query=q_dcat_classes, endpoint="https://www.w3.org/ns

# for row in query_r["results"]["bindings"]:
#     print(row)
```

```
/home/andre/Documents/Projects/SSHOC/repos/dcat-sandbox/venv/lib/python3.12/
site-packages/SPARQLWrapper/Wrapper.py:778: RuntimeWarning: Sending Accept h
eader '*/*' because unexpected returned format 'turtle' in a 'SELECT' SPARQL
query form
  warnings.warn(
```

In [41]:

```
q_dcat_classes =  prefixes + '''
SELECT *
WHERE {
    ?class a rdfs:Class ;
           rdfs:label ?label .
        #    dct:description ?desc .
    FILTER( LANG(?label) = 'en')
    # FILTER( LANG(?desc) = 'en')
}
'''

query_r = dcat_g.query(q_dcat_classes)
for r in query_r:
    print(r)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[41], line 2
      1 dir(query_r)
----> 2 query_r["results"]
      4 # print(str(construct_results, 'utf-8'))

File ~/Documents/Projects/SSHOC/repos/dcat-sandbox/venv/lib/python3.12/site-
packages/rdflib/graph.py:708, in Graph.__getitem__(self, item)
    705     return self.predicate_objects(item)  # type: ignore[arg-type]
    707 else:
--> 708     raise TypeError(
    709         "You can only index a graph by a single rdflib term or path,
or a slice of rdflib terms."
    710     )

TypeError: You can only index a graph by a single rdflib term or path, or a
slice of rdflib terms.
```